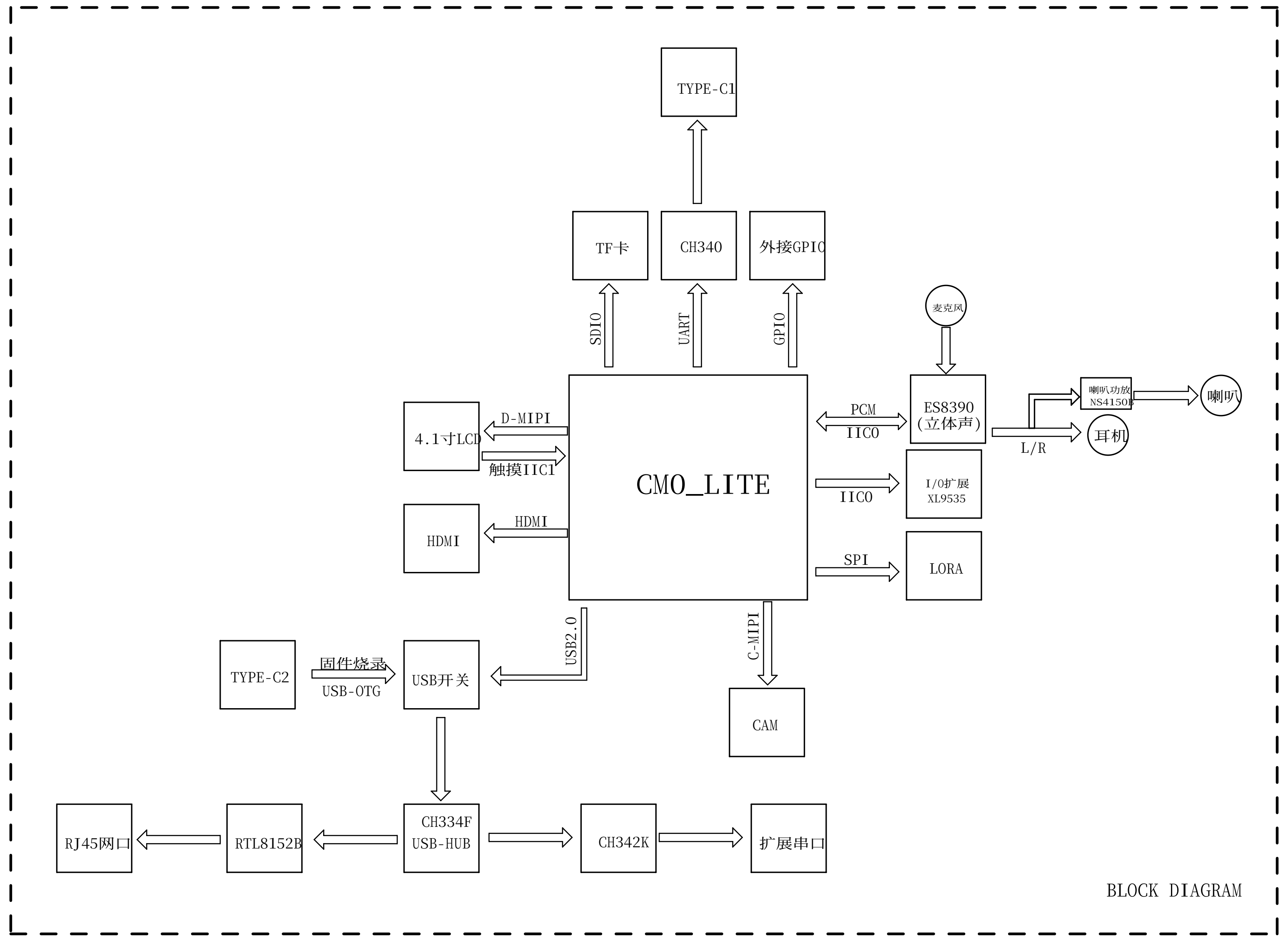
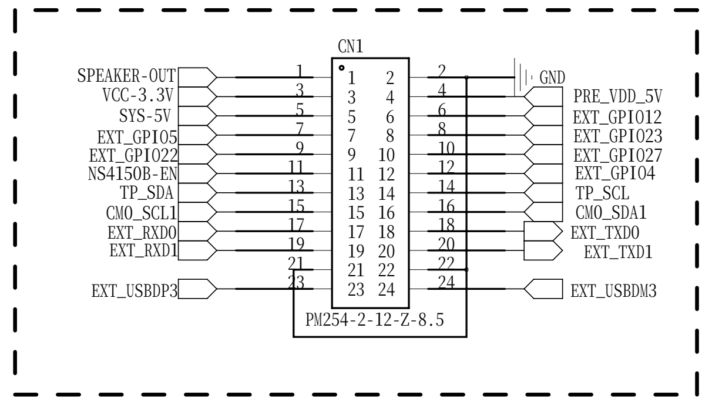
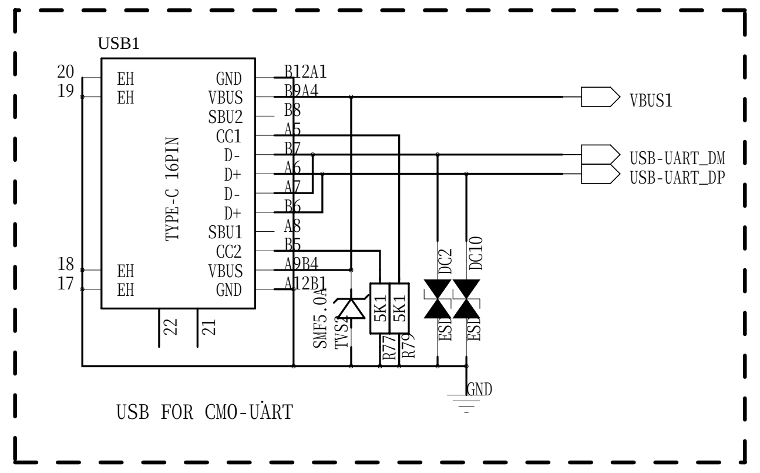
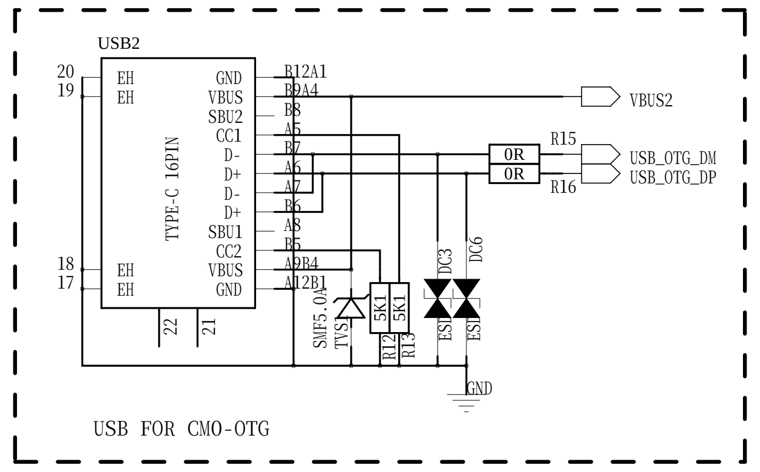

TYPE-C1
TF卡
CH340
外接GPIO
SDIO
UART
GPIO
麦克风
4.1寸LCD
D-MIPI
触摸IIC1
CMO_LITE
PCM
IIC0
ES8390
(立体声)
喇叭功放
NS4150B
喇叭
耳机
L/R
HDMI
HDMI
IIC0
I/O扩展
XL9535
SPI
LORA
USB2.0
C-MIPI
TYPE-C2
固件烧录
USB-OTG
USB开关
CAM
RJ45网口
RTL8152B
CH334F
USB-HUB
CH342K
扩展串口
BLOCK DIAGRAM

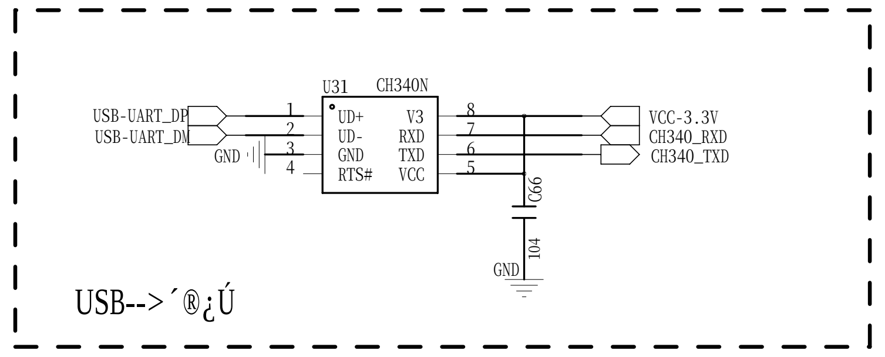




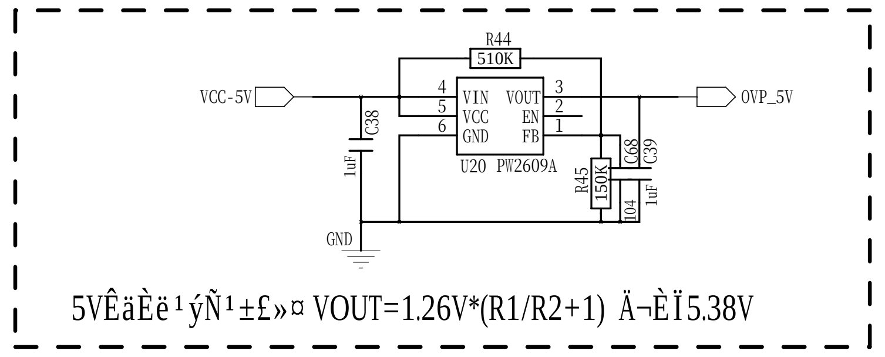
USB FOR CM0-UART



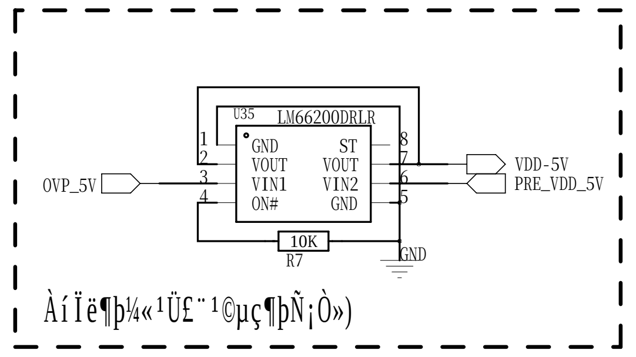
USB FOR CM0-OTG



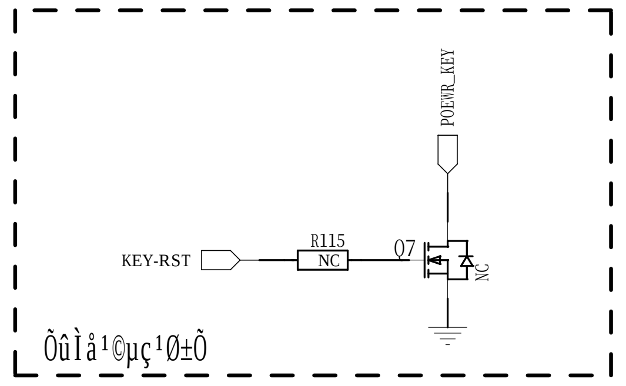
USB-->´®¿Ú



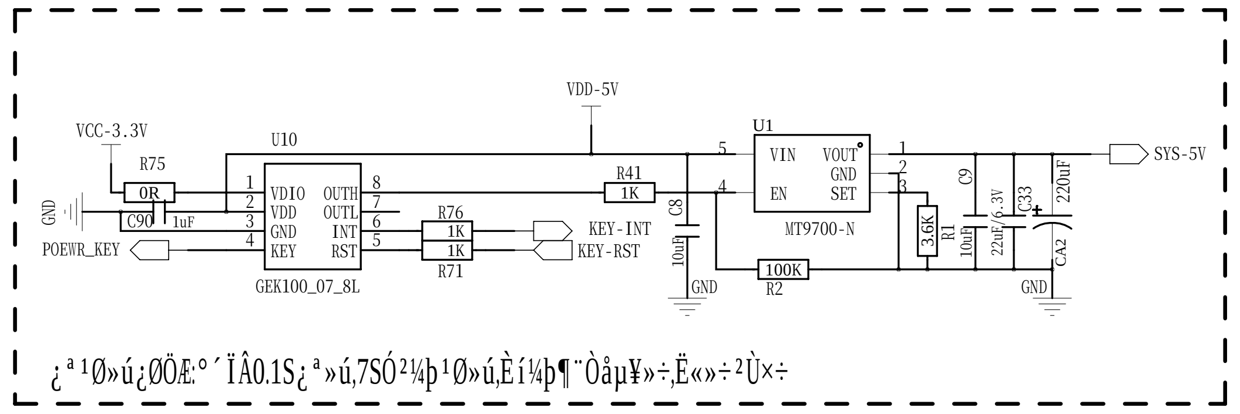
5VÊäÈë¹ýÑ¹±£»¤ VOUT=1.26V*(R1/R2+1) Ä¬ÈÏ5.38V



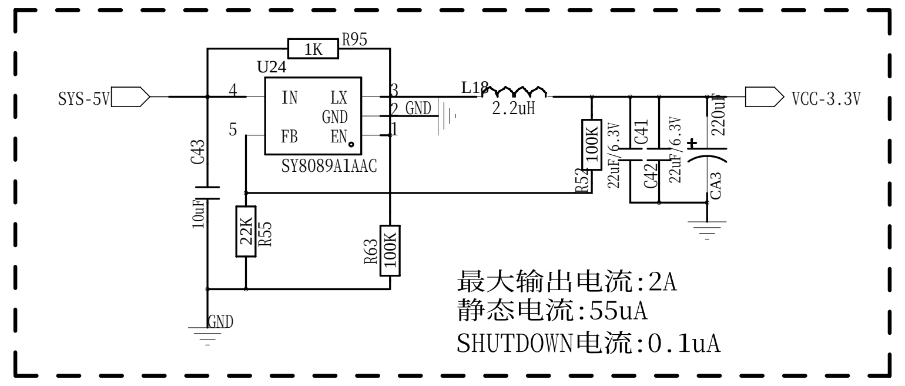
ÀíÏë¶þ¼«¹Ü£¨¹©µç¶þÑ¡Ò»)



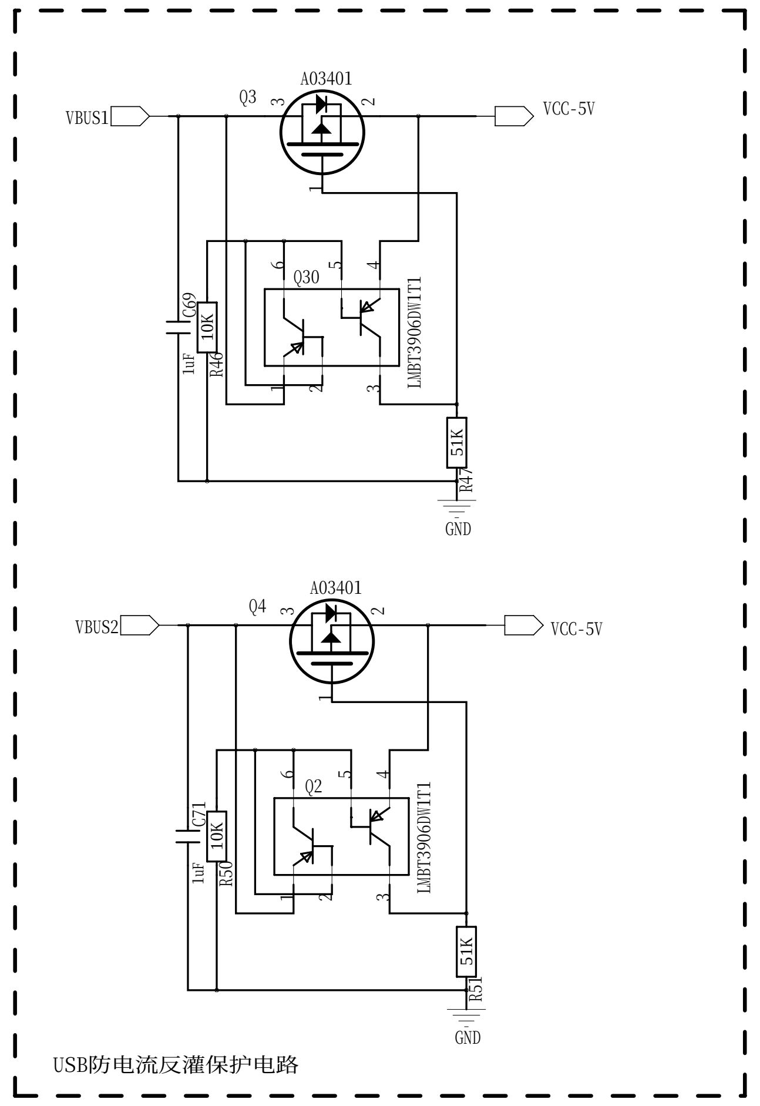
ÕûÌå¹©µç¹Ø±Õ



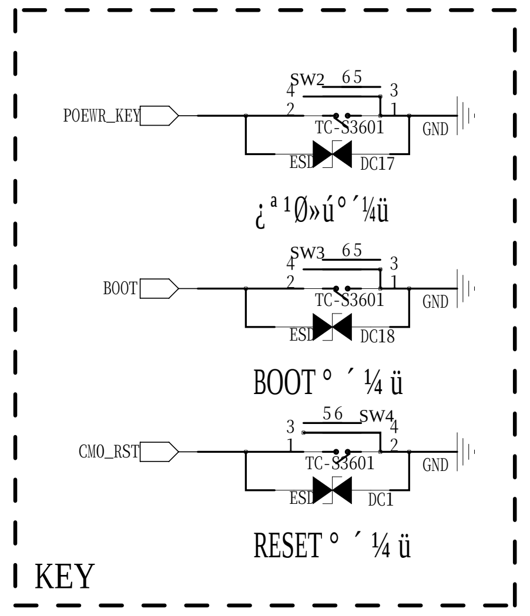
¿ª¹Ø»ú¿ØÖÆ:°´ÏÂ0.1S¿ª»ú,7SÓ²¼þ¹Ø»ú,Èí¼þ¶¨Òåµ¥»÷,Ë«»÷²Ù×÷



最大输出电流:2A
静态电流:55uA
SHUTDOWN电流:0.1uA



USB防电流反灌保护电路

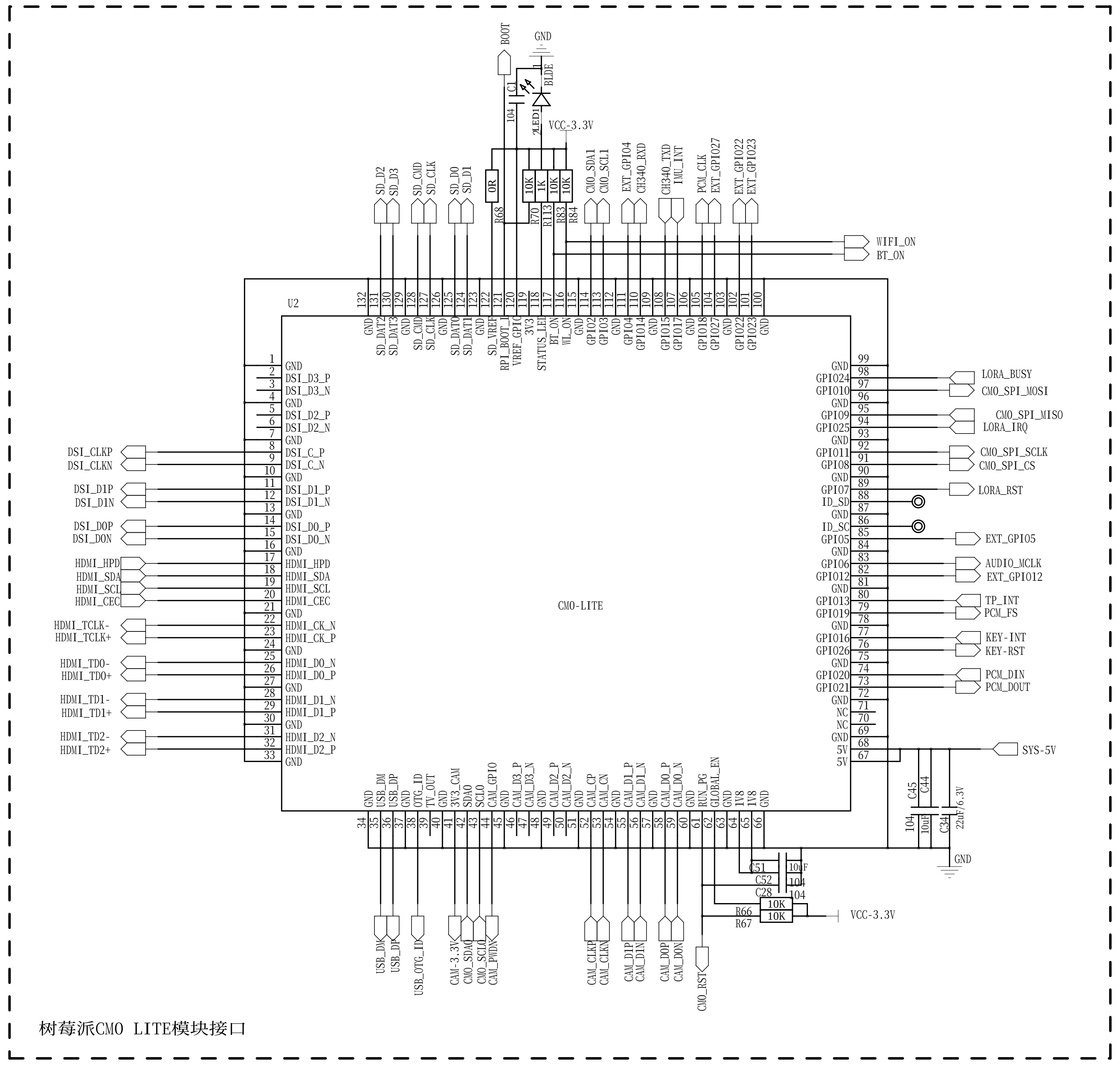


¿ª¹Ø»ú°´¼ü

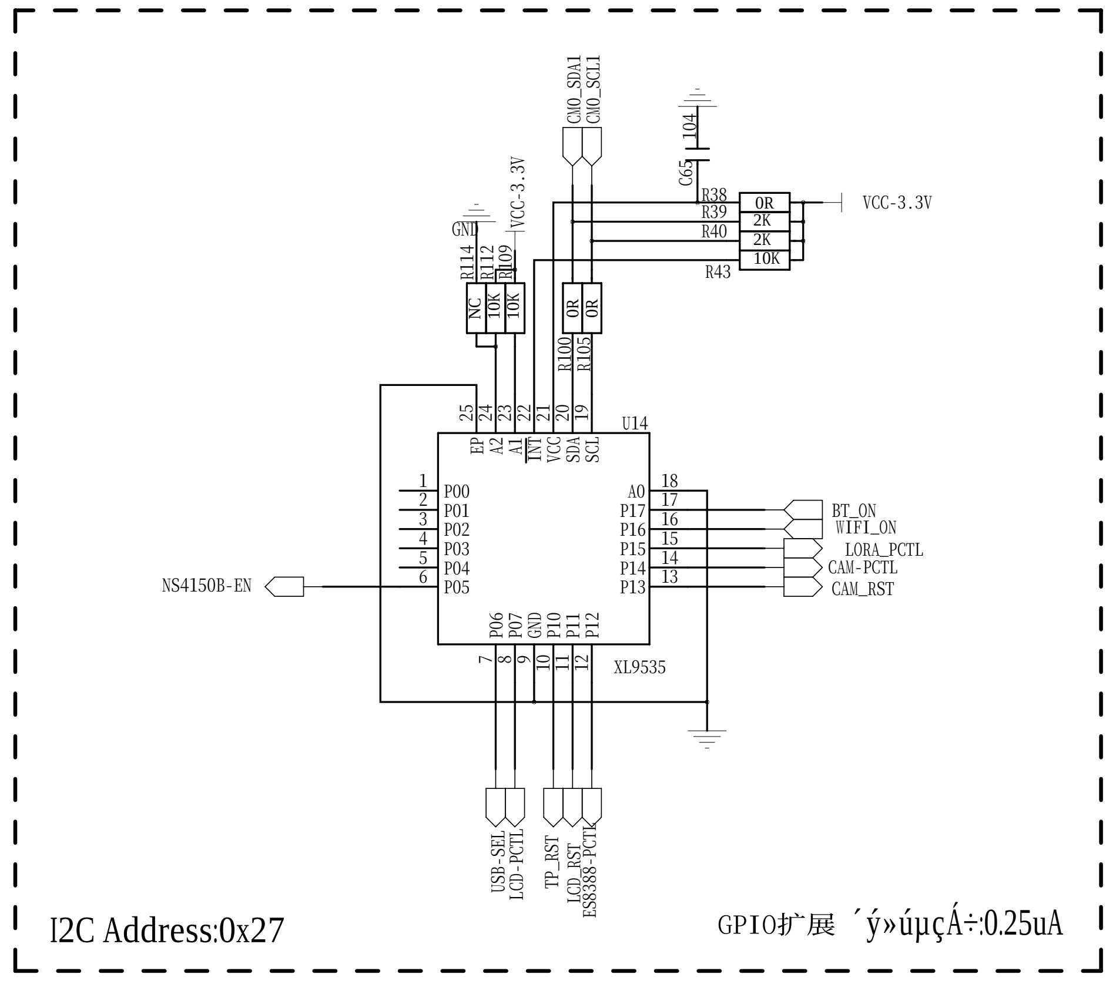
BOOT°´¼ü

RESET°´¼ü

KEY

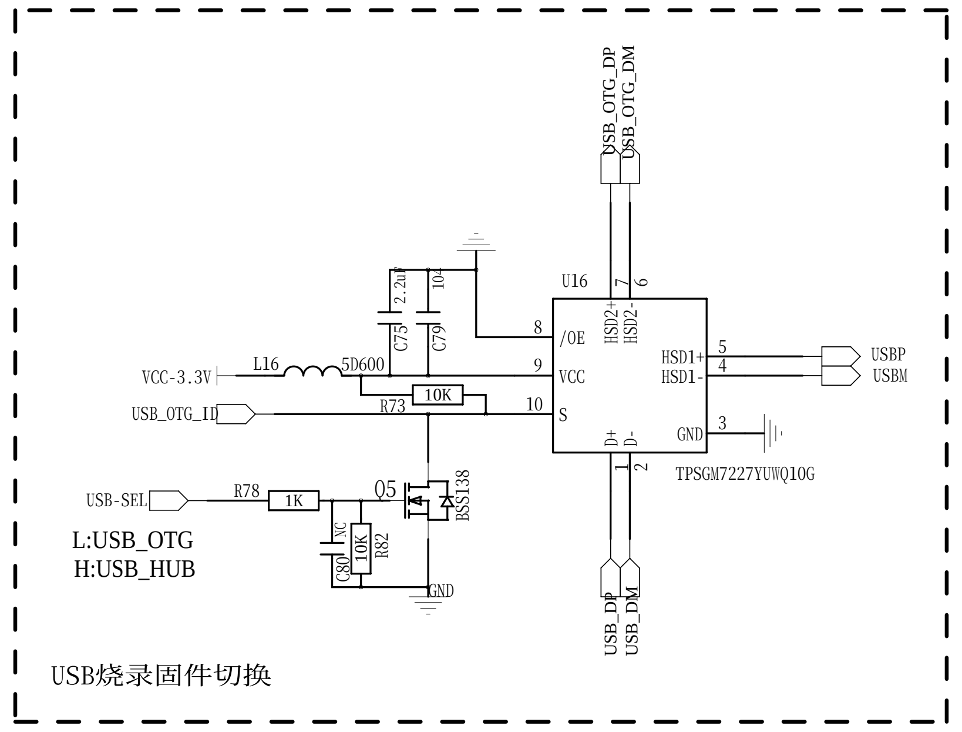


树莓派CM0 LITE模块接口

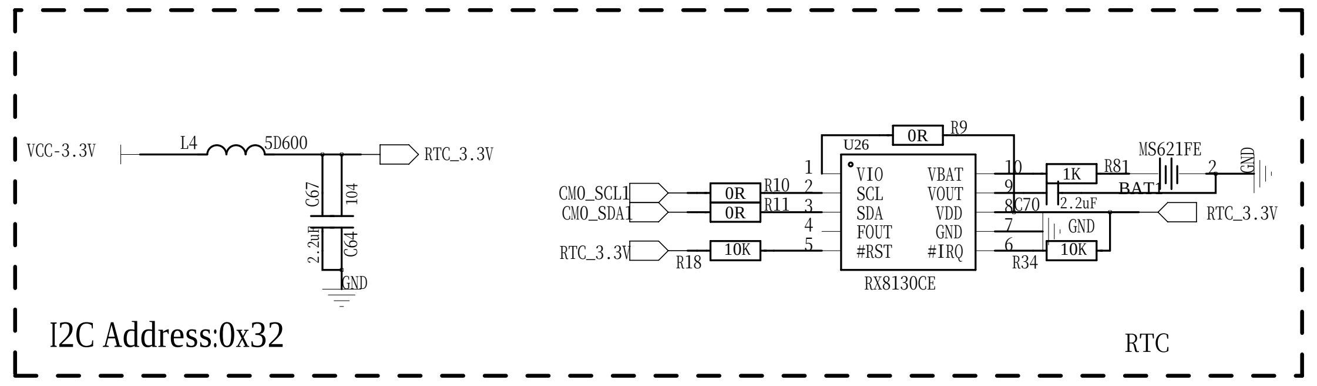


I2C Address:0x27

GPIO扩展 ´ý»úµçÁ÷:0.25uA

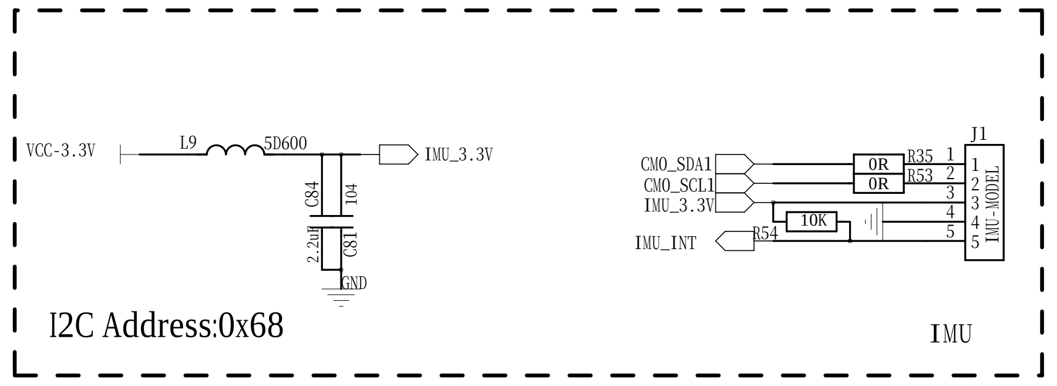


USB烧录固件切换



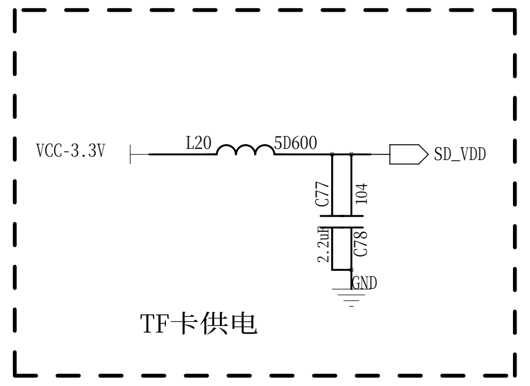
I2C Address:0x32

RTC

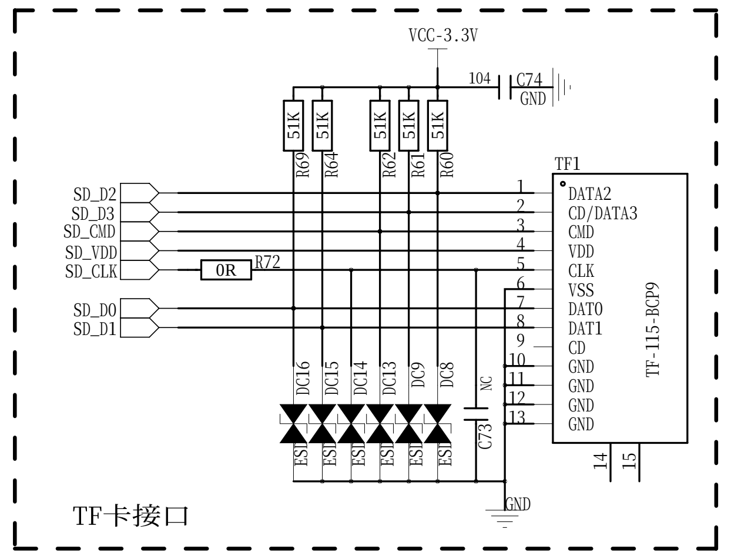


I2C Address:0x68

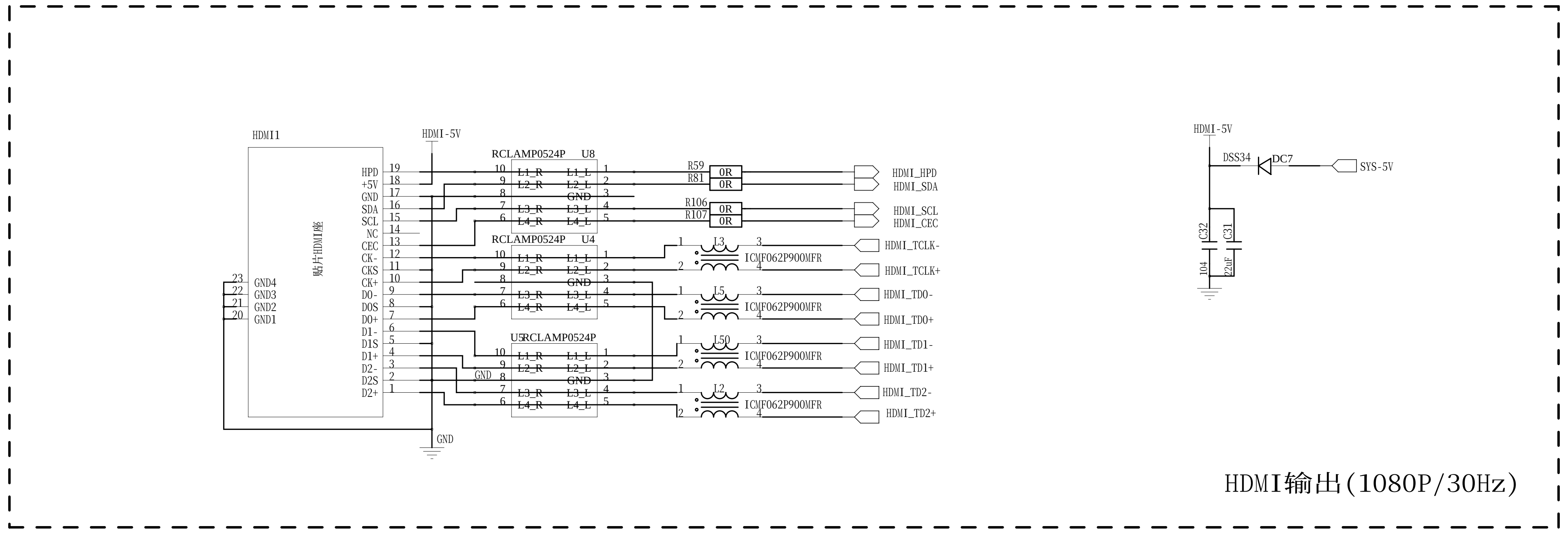
IMU



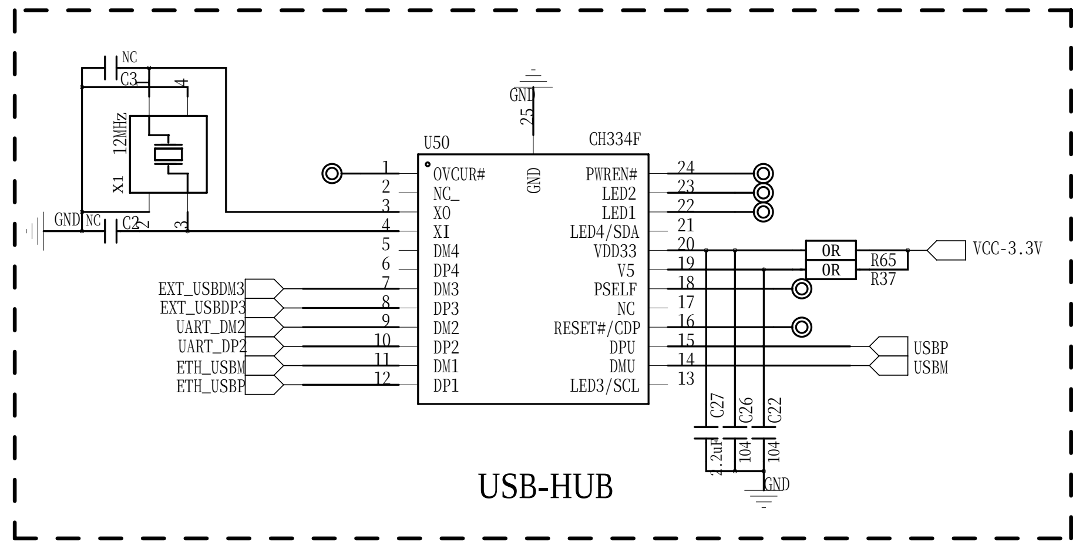
TF卡供电

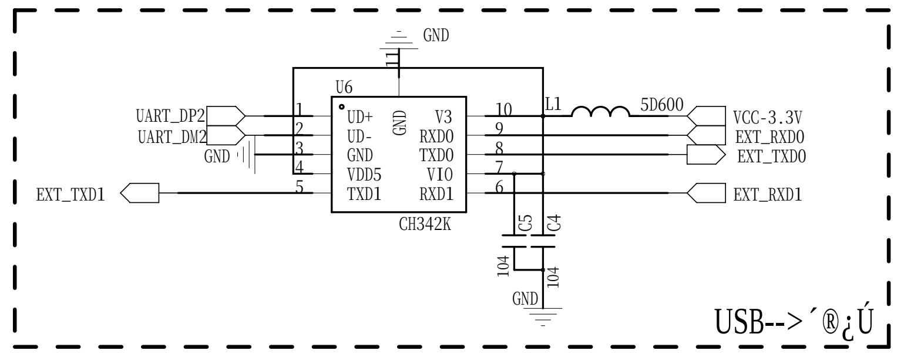


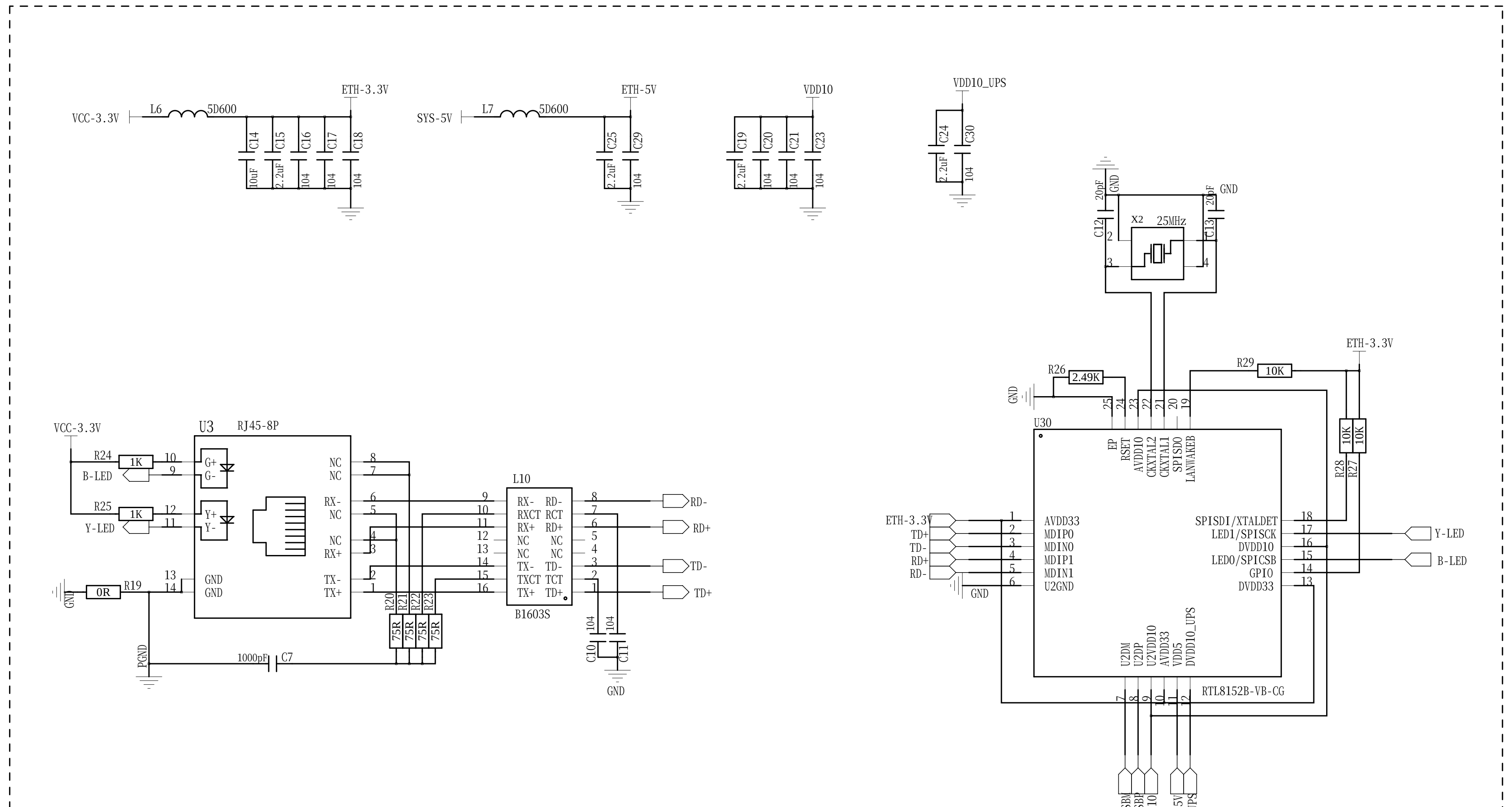
TF卡接口



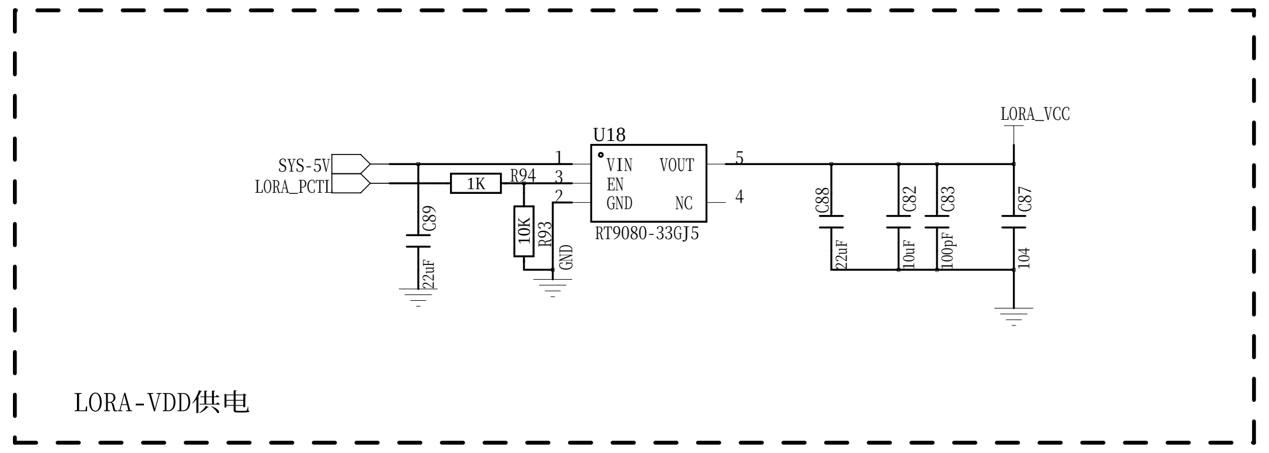
HDMI输出(1080P/30Hz)

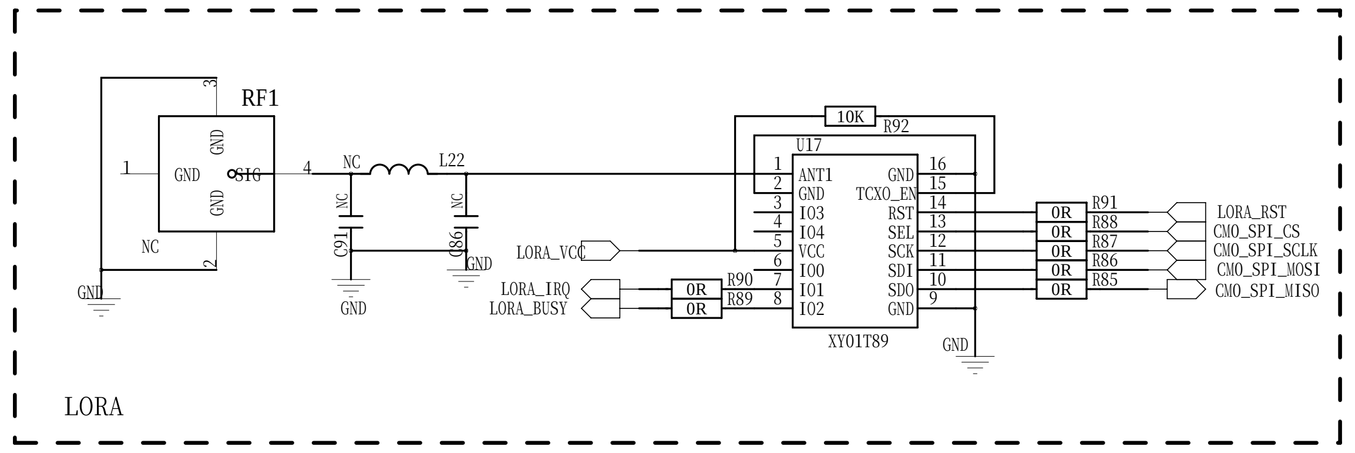

USB-HUB
U50
CH334F
GND
25
OVCUR#
NC_
XO
XI
DM4
DP4
DM3
DP3
DM2
DP2
DM1
DP1
PWREN#
LED2
LED1
LED4/SDA
VDD33
V5
PSELF
NC
RESET#/CDP
DPU
DMU
LED3/SCL
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
X1
12MHz
C3
NC
C2
NC
GND
EXT_USBDM3
EXT_USBDP3
UART_DM2
UART_DP2
ETH_USBM
ETH_USBP
0R
0R
R65
R37
VCC-3.3V
USBP
USBM
C27
C26
C22
2.2uF
104
104
GND

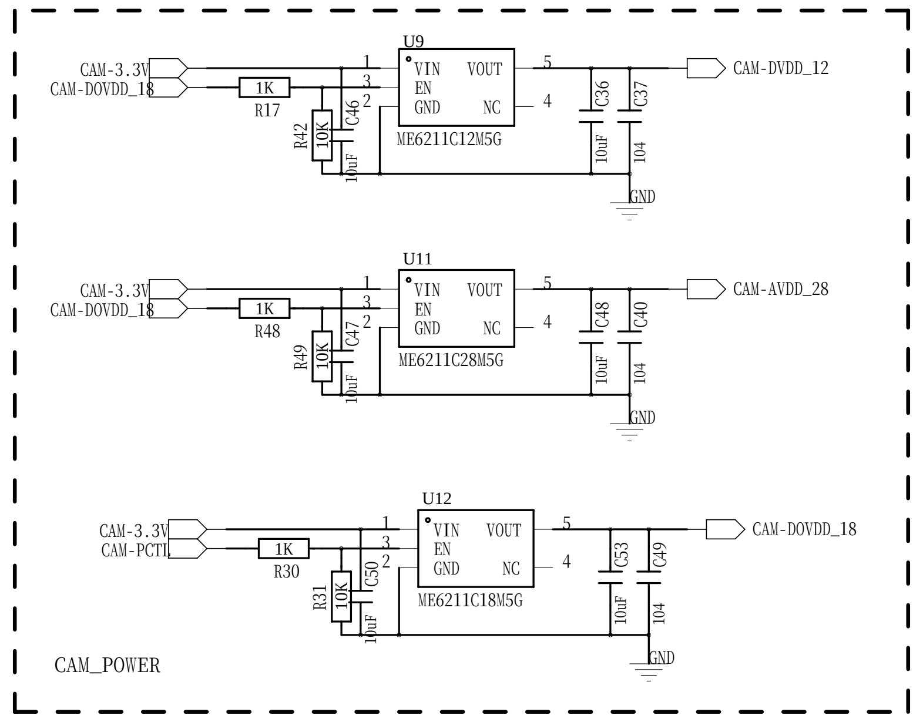

USB-->´®¿Ú
U6
CH342K
GND
11
UD+
UD-
GND
VDD5
TXD1
V3
RXD0
TXD0
VIO
RXD1
1
2
3
4
5
6
7
8
9
10
UART_DP2
UART_DM2
GND
EXT_TXD1
L1
5D600
VCC-3.3V
EXT_RXD0
EXT_TXD0
EXT_RXD1
C5
C4
104
104
GND



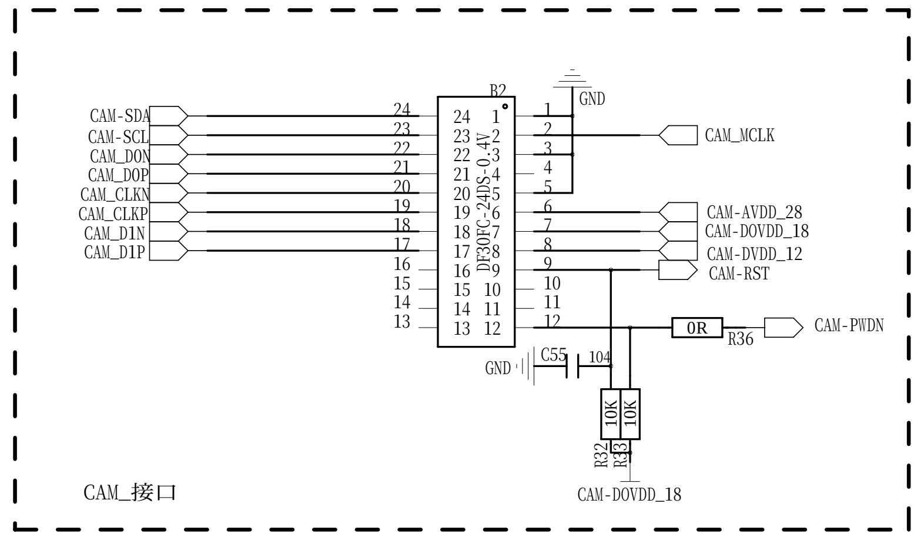
USB-->ETH(ÒÔÌ«Íø)

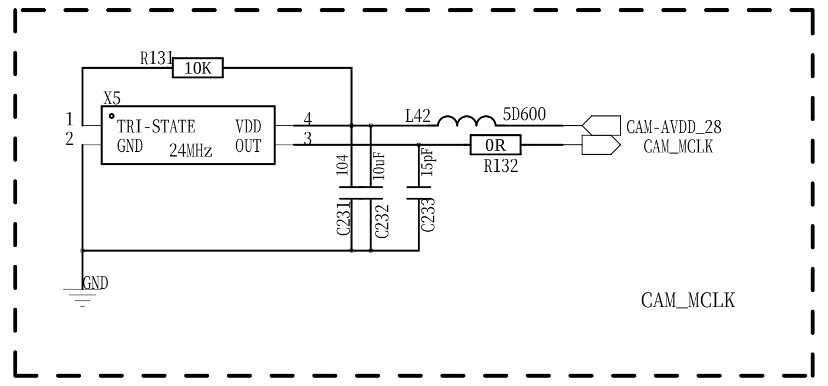

SYS-5V
LORA_PCTL
1K
R94
10K
R93
GND
C89
22uF
U18
VIN
EN
GND
VOUT
NC
1
3
2
5
4
RT9080-33GJ5
LORA_VCC
C88
22uF
C82
10uF
C83
100pF
C87
104
LORA-VDD供电

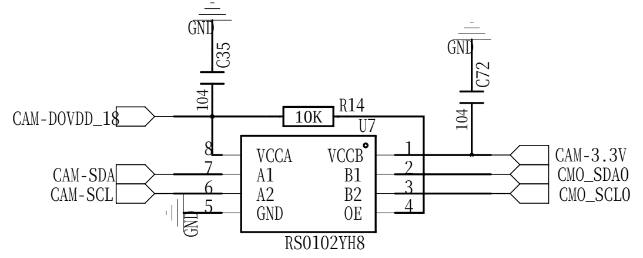

RF1
1
GND
SIG
GND
GND
3
2
NC
GND
4
NC
L22
NC
C91
NC
C86
GND
GND
10K
R92
U17
ANT1
GND
IO3
IO4
VCC
IO0
IO1
IO2
GND
TCXO_EN
RST
SEL
SCK
SDI
SDO
GND
1
2
3
4
5
6
7
8
16
15
14
13
12
11
10
9
XY01T89
LORA_VCC
LORA_IRQ
LORA_BUSY
0R
R90
0R
R89
0R
R91
0R
R88
0R
R87
0R
R86
0R
R85
LORA_RST
CM0_SPI_CS
CM0_SPI_SCLK
CM0_SPI_MOSI
CM0_SPI_MISO
GND
LORA

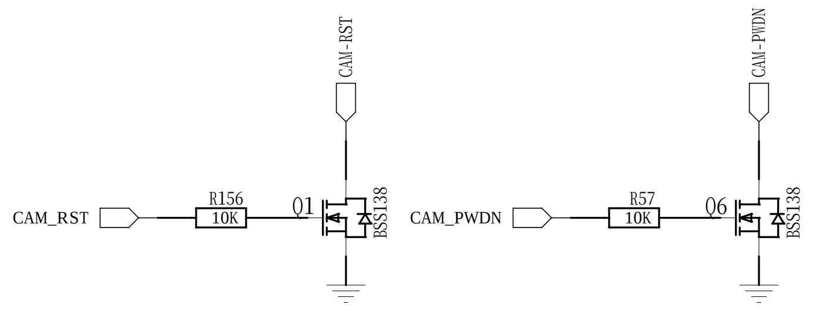


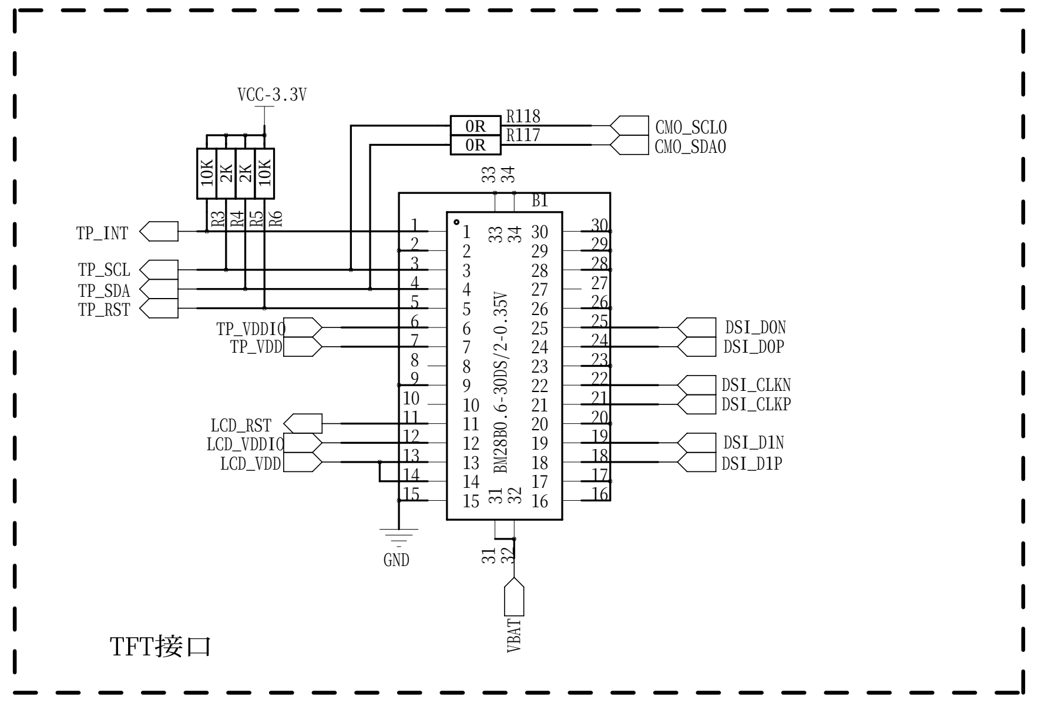
1:SDA	24:GND
2:SCL	23:MCLK
3:MDN0	22:GND
4:MDP0	21:PCLK
5:MCN	20:GND
6:MCP	19:AVDD
7:MDN1	18:DOVDD
8:MDP1	17:DVDD
9:NC	16:RESET
10:NC	15:VSYNC
11:NC	14:HREF
12:NC	13:PWDN



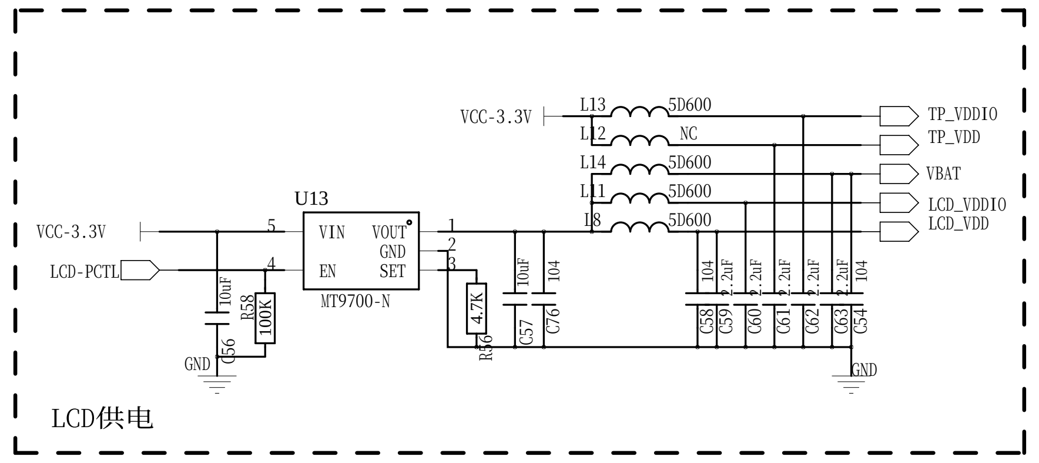




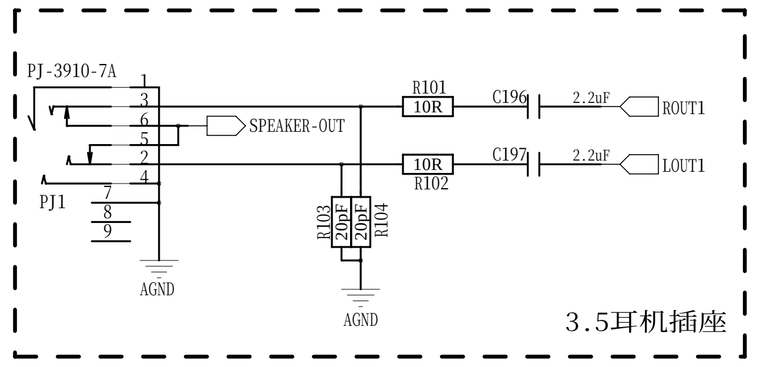



VCC-3.3V
10K
2K
2K
10K
R3
R4
R5
R6
0R
R118
0R
R117
CMO_SCL0
CMO_SDA0
33
34
B1
TP_INT
TP_SCL
TP_SDA
TP_RST
TP_VDDIO
TP_VDD
LCD_RST
LCD_VDDIO
LCD_VDD
BM28B0.6-30DS/2-0.35V
DSI_D0N
DSI_D0P
DSI_CLKN
DSI_CLKP
DSI_D1N
DSI_D1P
GND
31
32
VBAT

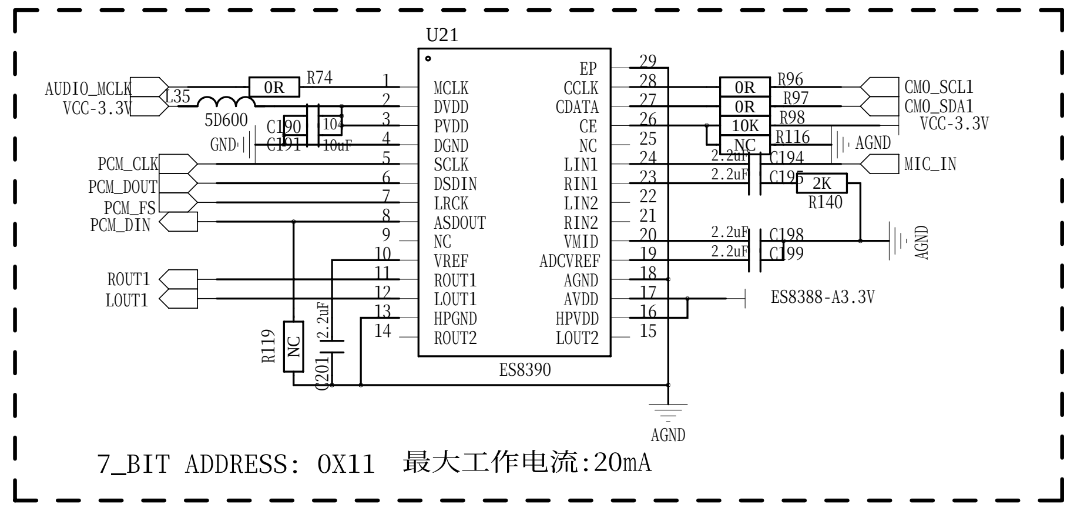
TFT接口


VCC-3.3V
LCD-PCTL
U13
VIN
EN
VOUT
GND
SET
MT9700-N
10uF
C56
R58
100K
GND
4.7K
R56
10uF
C57
104
C76
VCC-3.3V
L13
5D600
L12
NC
L14
5D600
L11
5D600
L8
5D600
104
C58
2.2uF
C59
2.2uF
C60
2.2uF
C61
2.2uF
C62
2.2uF
C63
104
C54
TP_VDDIO
TP_VDD
VBAT
LCD_VDDIO
LCD_VDD
GND

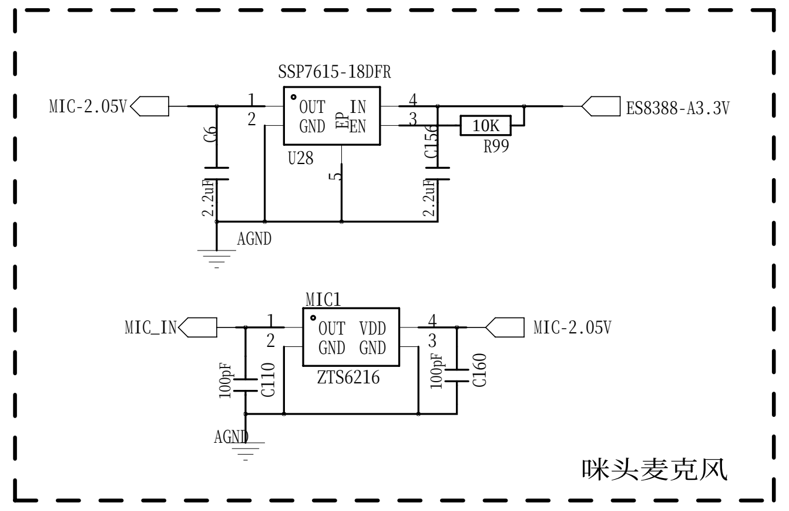
LCD供电



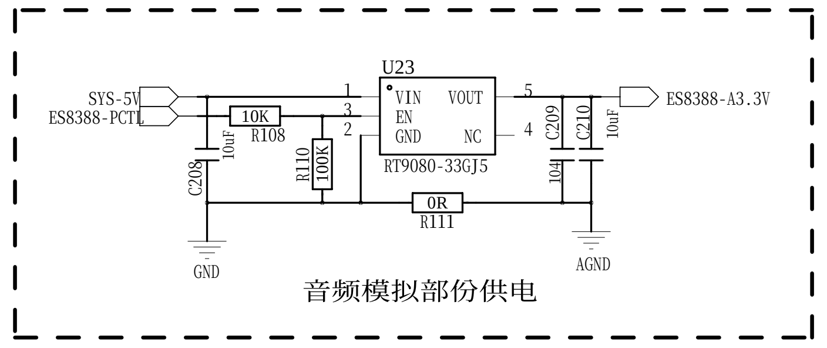
3.5耳机插座



7_BIT ADDRESS: 0X11 最大工作电流:20mA



咪头麦克风



音频模拟部份供电